

CallShield AI - Technical Documentation

Comprehensive Architecture, Technology Stack, Folder Structure & Log Reference

1. Official Project Links & Access

Resource	URL / Link
Live Web App (Vercel)	https://scam-audio-full-project.vercel.app
GitHub Repository	https://github.com/manav3009/scam-audio-detection
Android APK Package	CallShield_AI_v1.0.apk (Available on Desktop & Downloads)

2. Technology Stack & Frameworks

Layer	Technology	Description & Role
Backend Web Framework	Python 3.10+ & Flask 2.3	Core server framework handling routes, API endpoints, and sessions.
Frontend UI / UX	HTML5, CSS3, JavaScript	Responsive Glassmorphism interface for Mobile, Tablet, & Desktop viewports.
Speech-to-Text Engine	Web Speech API	Real-time browser-native voice recognition for incoming call & mic audio.
Audio Processing	Web Audio API & MediaRecorder	Live waveform visualization and binary audio chunk recording.
Scam Detection AI Engine	Custom Heuristic NLP Engine	Analyzes keywords, risk weights, urgency patterns, & banking threats.
AI Cybersecurity Chatbot	DuckDuckGo Search Engine API	Fetches real-time web security data and safety responses.
Android Native Layer	Java & Android SDK (API 21-34)	Native hardware permissions (mic, call state listener) & Android WebView.
Cloud Infrastructure	Vercel Serverless Hosting	Production cloud hosting with automated CI/CD deployment.
Build System	Gradle 8.13	Compiles and packages standalone standalone CallShield_AI_v1.0.apk.

3. Directory & Folder Structure

A. Web Application Structure:

```
Scam_audio_Full_Project/ (Web Application)
├── api/
│   ├── index.py           # Vercel serverless deployment entry point
│   └── core/
│       ├── detector.py    # Fraud NLP engine & scoring logic (0-100%)
│       ├── chatbot_data.py # Cybersecurity knowledge base & search query engine
│       ├── test_data.py   # Sample scam transcripts & saved live audio calls
│       └── routes/
│           └── analysis_routes.py # Audio upload, live mic streaming & analysis APIs
```

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■   ■■■ auth_routes.py           # Admin/Student authentication routes
■   ■■■ chatbot_routes.py        # AI assistant conversational API endpoints
■   ■■■ views_routes.py          # Page render routes (Index, Dashboard, Login)
■■■ static/
■   ■■■ css/                     # Custom stylesheets (Glassmorphism layout)
■   ■■■ js/                      # Real-time Web Speech API & audio visualizer
■■■ templates/
■   ■■■ base.html                # Master UI layout with responsive navigation
■   ■■■ index.html               # Landing page & quick scanner
■   ■■■ login.html               # Auth portal for Students & Admin
■   ■■■ dashboard.html           # Student & Admin analytics dashboard
■   ■■■ modules/
■       ■■■ live_call.html        # Real-time live call detector & waveform streaming
■       ■■■ recorded.html         # Offline audio file uploader & speech analyzer
■       ■■■ chatbot.html          # Cybersecurity AI assistant interface
■       ■■■ awareness.html        # Scam awareness guide & preventive safety tips
■■■ app.py                       # Flask application factory & session setup
■■■ requirements.txt              # Python dependencies (Flask, requests, duckduckgo-search)
■■■ vercel.json                  # Vercel deployment & URL rewriting rules

```

B. Android Application Structure:

```

CallShield_Android_App/ (Native Android Project)
■■■ app/
■   ■■■ build.gradle              # SDK config (minSdk 21 / targetSdk 34) & dependencies
■   ■■■ src/
■       ■■■ main/
■           ■■■ AndroidManifest.xml # Permissions (RECORD_AUDIO, INTERNET, CALL_STATE)
■           ■■■ java/com/callshield/scamaudiodetector/
■               ■■■ MainActivity.java # WebChromeClient, Android WebView & Telephony
listener
■       ■■■ res/
■           ■■■ drawable/         # Application launch graphics & icons
■■■ build.gradle                  # Project-level Gradle build configuration
■■■ gradle.properties             # JVM environment & Gradle 8.13 build settings

```

4. Logs & History Storage Reference

Category	Storage Location	Details & Persistence
Chatbot History	Browser localStorage (Key: callshield_chat_history)	Stored client-side. Persists across page reloads and device restarts.
Live Call Recordings	Server Memory / TEST_AUDIO_FILES in core/test_data.py	Saved live calls automatically populate the 'Recorded Audio' dropdown.
Server Web Logs	Vercel Dashboard Functions & Terminal stdout/stderr	Real-time HTTP request logs, status codes, and server analytics.
Android Native Logs	Android System Logcat (Tag: CallShield_App)	Captures microphone permissions, call state changes, and WebView errors.